

## RIO DE JANEIRO SANTA CRUZ, Brazil

WMO: 837410

Lat: 22.932S

Lon: 43.719W

Elev: 3

StdP: 101.29

Time Zone: -3.00 (SAB)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 7             | 13.8       | 14.9 | 10.2                          | 7.7 | 21.2 | 11.8 | 8.6 | 20.8 | 9.9                   | 23.8 | 9.0 | 23.6 | 1.9                   | 40   | 0.319 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 2             | 9.7                    | 36.2            | 24.9 | 35.1 | 24.7 | 33.9 | 24.7 | 27.1                | 32.9 | 26.6 | 32.0 | 26.0 | 31.2 | 4.5                  | 20   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 25.2                            | 20.4 | 30.4 | 25.0 | 20.1 | 30.1 | 24.2 | 19.2 | 28.7 | 86.0          | 33.5 | 83.0 | 32.1 | 80.5 | 31.2 | 30.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 8.9               | 8.0  | 7.0 | 11.4                       | 39.3 | 1.4                | 0.9 | 10.4                                               | 40.0 | 9.5        | 40.5 | 8.7        | 41.0 | 7.7        | 41.6 |
|                   |      |     | 10.6                       | 27.9 | 1.4                | 1.1 | 9.6                                                | 28.7 | 8.8        | 29.4 | 8.1        | 30.0 | 7.1        | 30.9 |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 24.4   | 27.6  | 28.2  | 26.6  | 25.0  | 22.8  | 21.7  | 21.6  | 22.2  | 23.0  | 23.8  | 24.5  | 26.4 |
|                                                                               | DBStd       | 3.29   | 2.55  | 2.08  | 2.16  | 2.19  | 2.27  | 2.34  | 2.54  | 2.57  | 2.83  | 2.62  | 2.64  | 2.42 |
|                                                                               | HDD10.0     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                               | HDD18.3     | 9      | 0     | 0     | 0     | 0     | 0     | 2     | 4     | 2     | 1     | 0     | 0     | 0    |
|                                                                               | CDD10.0     | 5269   | 546   | 511   | 513   | 449   | 397   | 352   | 361   | 377   | 391   | 429   | 437   | 508  |
|                                                                               | CDD18.3     | 2237   | 288   | 277   | 255   | 199   | 138   | 105   | 107   | 120   | 141   | 171   | 187   | 249  |
|                                                                               | CDH23.3     | 17077  | 2885  | 2862  | 2119  | 1422  | 756   | 526   | 610   | 727   | 838   | 1004  | 1230  | 2099 |
|                                                                               | CDH26.7     | 6482   | 1272  | 1297  | 754   | 456   | 202   | 133   | 177   | 257   | 324   | 362   | 429   | 819  |
| Wind                                                                          | WSAvg       | 3.1    | 3.1   | 3.1   | 2.7   | 2.7   | 2.6   | 2.7   | 3.1   | 3.4   | 3.6   | 3.5   | 3.3   | 3.2  |
| Precipitation                                                                 | PrecAvg     | 1414   | 246   | 161   | 181   | 86    | 56    | 33    | 29    | 31    | 72    | 107   | 181   | 232  |
|                                                                               | PrecMax     | 1957   | 392   | 343   | 349   | 189   | 136   | 81    | 79    | 82    | 155   | 215   | 294   | 455  |
|                                                                               | PrecMin     | 752    | 95    | 29    | 84    | 23    | 20    | 4     | 3     | 5     | 13    | 45    | 85    | 116  |
|                                                                               | PrecStd     | 256    | 80    | 79    | 79    | 43    | 28    | 19    | 20    | 18    | 35    | 51    | 59    | 75   |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 37.2  | 37.8  | 36.0  | 34.1  | 32.9  | 32.0  | 32.2  | 34.2  | 36.0  | 36.0  | 36.0  | 37.0 |
|                                                                               |             | MCWB   | 25.1  | 25.0  | 25.9  | 25.9  | 24.5  | 22.7  | 21.1  | 21.6  | 22.4  | 23.9  | 24.7  | 25.2 |
|                                                                               | 2%          | DB     | 35.9  | 36.1  | 34.0  | 32.8  | 30.9  | 30.0  | 30.2  | 31.9  | 33.1  | 33.0  | 33.2  | 34.9 |
|                                                                               |             | MCWB   | 25.3  | 24.9  | 25.4  | 24.7  | 23.1  | 21.7  | 21.2  | 21.1  | 21.7  | 23.3  | 24.1  | 24.7 |
|                                                                               | 5%          | DB     | 34.1  | 34.8  | 32.1  | 30.9  | 29.0  | 28.1  | 28.9  | 29.8  | 30.2  | 30.2  | 31.0  | 33.0 |
|                                                                               |             | MCWB   | 25.3  | 24.9  | 25.0  | 24.4  | 22.4  | 21.1  | 20.8  | 20.6  | 21.6  | 22.7  | 23.5  | 25.0 |
|                                                                               | 10%         | DB     | 32.8  | 33.1  | 30.8  | 29.1  | 27.1  | 26.2  | 26.9  | 27.2  | 27.9  | 28.1  | 29.0  | 31.0 |
|                                                                               |             | MCWB   | 25.1  | 25.1  | 25.1  | 23.8  | 21.8  | 20.7  | 20.4  | 20.5  | 21.3  | 22.3  | 23.2  | 24.8 |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 27.7  | 28.3  | 27.3  | 27.0  | 25.4  | 24.1  | 23.9  | 24.4  | 25.2  | 26.0  | 26.1  | 27.7 |
|                                                                               |             | MCDB   | 33.9  | 35.2  | 32.8  | 32.5  | 30.6  | 29.9  | 29.3  | 30.5  | 32.8  | 33.1  | 32.4  | 33.3 |
|                                                                               | 2%          | WB     | 26.8  | 27.1  | 26.5  | 26.0  | 24.2  | 22.7  | 22.6  | 23.0  | 23.6  | 24.5  | 25.0  | 26.4 |
|                                                                               |             | MCDB   | 32.5  | 32.9  | 31.3  | 30.4  | 28.8  | 27.1  | 27.3  | 28.1  | 29.3  | 30.2  | 31.0  | 32.3 |
|                                                                               | 5%          | WB     | 26.3  | 26.4  | 25.8  | 25.0  | 23.3  | 22.1  | 21.7  | 22.1  | 22.7  | 23.6  | 24.2  | 25.5 |
|                                                                               |             | MCDB   | 31.7  | 31.8  | 30.2  | 28.9  | 27.1  | 26.0  | 26.0  | 26.8  | 27.0  | 28.2  | 29.4  | 30.9 |
|                                                                               | 10%         | WB     | 25.6  | 25.7  | 25.3  | 24.3  | 22.5  | 21.3  | 20.9  | 21.1  | 21.9  | 22.8  | 23.6  | 24.8 |
|                                                                               |             | MCDB   | 30.7  | 30.7  | 29.2  | 28.0  | 25.9  | 24.8  | 24.7  | 25.1  | 25.9  | 26.6  | 27.9  | 29.6 |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 9.0   | 9.7   | 7.9   | 8.6   | 8.9   | 9.1   | 9.5   | 9.7   | 9.0   | 8.5   | 7.8   | 8.3  |
|                                                                               |             | MCDBR  | 11.6  | 12.3  | 10.5  | 11.0  | 11.3  | 11.9  | 12.8  | 14.0  | 13.7  | 12.9  | 12.1  | 11.7 |
|                                                                               | 5% WB       | MCWBR  | 4.1   | 4.0   | 3.9   | 4.4   | 4.8   | 5.1   | 5.3   | 5.2   | 4.8   | 4.8   | 4.6   | 4.6  |
|                                                                               |             | MCWBR  | 9.9   | 10.3  | 9.1   | 9.5   | 9.8   | 9.9   | 10.7  | 11.4  | 10.8  | 10.7  | 10.1  | 9.8  |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.417  | 0.414 | 0.412 | 0.406 | 0.386 | 0.388 | 0.387 | 0.421 | 0.505 | 0.476 | 0.442 | 0.430 |      |
|                                                                               | taud        | 2.457  | 2.447 | 2.461 | 2.463 | 2.495 | 2.473 | 2.457 | 2.341 | 2.083 | 2.232 | 2.355 | 2.406 |      |
|                                                                               | Ebn at Noon | 929    | 918   | 893   | 855   | 832   | 807   | 821   | 826   | 795   | 853   | 900   | 918   |      |
|                                                                               | Edh at Noon | 121    | 120   | 113   | 104   | 93    | 92    | 96    | 116   | 162   | 147   | 133   | 127   |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 5.95   | 6.19  | 5.31  | 4.68  | 3.73  | 3.47  | 3.60  | 4.28  | 4.58  | 5.14  | 5.32  | 5.86  |      |
|                                                                               | RadStd      | 0.85   | 0.63  | 0.45  | 0.34  | 0.21  | 0.27  | 0.31  | 0.34  | 0.59  | 0.60  | 0.42  | 0.51  |      |

## Historical Trends

|                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| Station Only           | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| Regional (3 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
|                        | N/A   | N/A     | -1.69  | N/A     | -1.40 | -1.66 | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page